

An affine-relation obstruction to discretizing a second die

Methodological research note; independently reviewed

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Theorem 1. *Let A be supported on finitely many distinct points $a_1, \dots, a_K \in [0, 1]$, with positive rational probabilities $w_1, \dots, w_K$. Suppose $\sum_i w_i a_i = 1/2$ and that*

$$\left\{ c \in \mathbb{Q}^K : \sum_i c_i a_i \in \mathbb{Q} \right\} = \mathbb{Q}(w_1, \dots, w_K).$$

There is no finitely supported random variable B with rational probabilities such that the independent variables A, B, U , with U uniform on $[0, 1]$, have each of their six strict orders with probability $1/6$.

Proof. Put $b_i = \Pr(B > a_i) \in \mathbb{Q}$. The order $U < A < B$ would give

$$\sum_i w_i a_i b_i = \frac{1}{6}.$$

The affine-relation hypothesis then forces $(w_i b_i)_i$ to be a rational multiple of $(w_i)_i$. Thus all b_i have one common value b . The six strict order probabilities sum to one, so A and B have no ties. Marginalizing their relative order gives $b = \Pr(A < B) = 1/2$. But then

$$\Pr(U < A < B) = b \sum_i w_i a_i = \frac{1}{4},$$

contradicting $1/6$. □

For equiprobable A , the hypothesis says that the only rational coefficient combinations of its node values that are rational are constant multiples of their sum. This note does not claim that all real interval quadratures have this property, or that two small finite dice are impossible. It identifies an arithmetic obstruction to a construction strategy which first selects a real

scalar quadrature and then tries to discretize another variable while leaving the remaining variables exactly uniform. Special affine relations among the first quadrature's nodes are necessary for that strategy.